TourCount 3.8.5

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1. Introduction

TourCount should support you when recording European butterflies and day-flying moth (**Fig. 1**). It allows to register individuals species-specific, separated by gender and metamorphic stages as well as individually with condition and locality in the field. It offers prepared result outputs and substitutes your field book and pencil, and if applicable a camera for documentary pictures.



Fig. 1: Starting Page

Pro “tour” a new instance of a prepared database will be used. Database files are tour-, region- or biotope-related, thus Databases can be individually created and adapted regarding expected butterfly species.

The recorded data (metadata, counts and annotations) is prepared as a report for an easy manual transfer into a butterfly registration system, e.g. at **https://web.app.ufz.de/tagfalter-monitoring/** there under “SINGLE FINDINGS REPORT” or **https://www.falterfunde.de/platform/s4y/falterfunde/index.do**.
It can be read directly on the smartphone or exported as an MS Excel spreadsheet and transferred to a PC, where it can be viewed or processed more easily.

The app is published with source code and documentation on **https://github.com/wistein/TourCount**. It is open source, has neither tracking nor advertising functions and does not make use of Google Play Services.
It demands only for permits which are needed for its serviceable functions:

- **All files access** which is needed for importing databases and species lists and exporting databases, species lists and the counting results as .db, .csv or .xlsx files (exclusively in the “Documents/TourCount” folder and, if applicable, in “Documents/TransektCount,” if TransektCount is installed),
- optional **Location service** with GPS to get the exact location of each count, in conjunction with **Internet access** for **reverse geocoding** (to get location address data by the Nominatim service of OpenStreetMap) and
- optional **Wakelock** to hinder the app from being switched off by the system when inactive, but get dimmed and get an insensitive screen when pocketed or held near the body.

2. Setting up

For installation hints refer to **chapter 5**.

When starting the app for the first time, you must grant permissions for storage and precise location tracking. TourCount reads from and writes to only the public directory **Documents/TourCount** which is created when the app is first launched (and **Documents/TransektCount** if the complementary app TransektCount is installed, see paragraph **6.3**). This data directory will not be deleted when TourCount is uninstalled; the data stored there is retained.

Before initial use you should adapt the app settings (→ **4. Further Functions**).

The preconfigured species list should be adapted to the species to be expected regionally. Use the editing icons in the app bar of the counting page (**Fig. 2**):

- ⊕ "**Add Species**",
- 🗑 "**Remove Species**" or
- ✎ "**Edit Terms**"

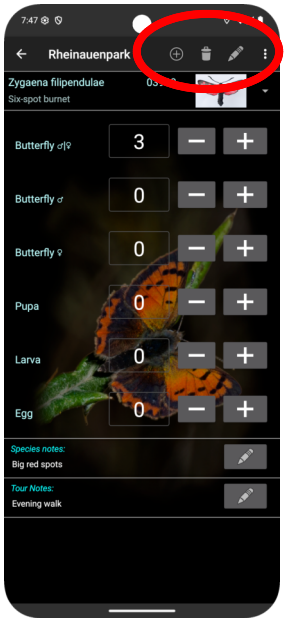


Fig. 2: Counting Page

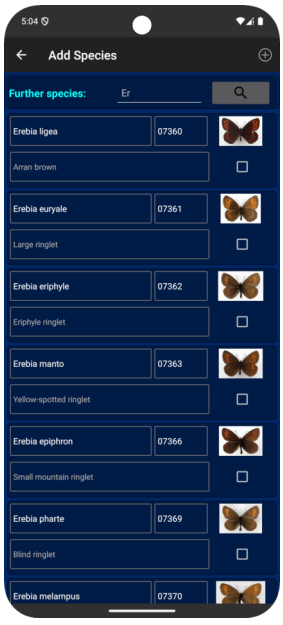


Fig. 3: "Add Species" page

All 3 editing pages offer a **preselection** to ease the selection of a distinguished species. Enter 2 or more letters of the scientific species name and then click the 🔍-button to limit the shown list. You may enter more letters and click the 🔍-button again to further reduce the selection list.

Under "**Add Species**" (**Fig. 3**), further species can be added from the larger integrated list of European species using the checkboxes. (The added species then disappear from the selection list).

A placeholder for an indeterminate species (NN) can be added from the end of the "**Add Species**" scroll-down list, which can be edited under "**Edit terms**" once it has been defined.

Under "**Remove Species**" mark all irrelevant species and remove them with the 🗑 button.

Under "**Edit Terms**" if necessary, adjust tour name, comment, species name and code. (Species names, scientific and English and species code, 5 digits, with leading zeros).

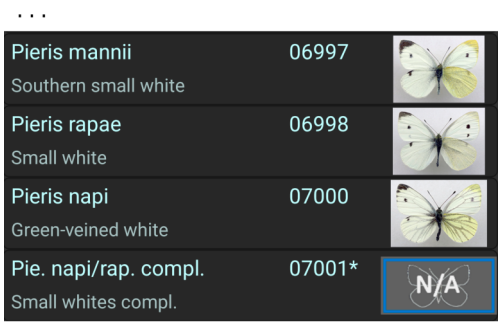


Fig. 4: Species list (partly)

The unique codes will be used as a reference to show corresponding butterfly pictures on the counting and results page and as an option to sort the list. The codes derive from the numbering scheme of European butterflies by Karsholt/Razowski, as used e.g. in the German Lepiforum (https://lepiforum.org/).

The *-sign attached to code 07001 in **Fig. 4** indicates a complex of species that are difficult to distinguish. For sorting purposes, the largest code within such species complexes should be used for the group.

Caution: An incorrect code will display an "N/A" or incorrect image. If necessary, see the list in 5 languages "**List of Coded Butterflies for TourCount and TransektCount.pdf**" on https://github.com/wistein/TourCount/tree/master/docs.

Changes are applied using the corresponding ⊕, 🗑 or 🖫 function buttons.

Alternatively, a more comprehensive or specialized sample **species list** with local species names in German, English, French, Italian or Spanish can be downloaded from https://github.com/wistein/

TourCount/tree/master/docs, copied to the app's data directory “**Documents/TourCount**” and imported from there, and then further customized.

Then you may enter the observer name on the “**Edit Metadata**” page (**Fig. 5**) and finish this by clicking the saving symbol. If **reverse geocoding** is activated under “**Settings**” location-related metadata will be derived from GPS coordinates and will be inserted automatically. The editable metadata may also be modified anytime later.

When you have created the counting list for all expected species and entered the basic metadata, the internal database is ready for export as a loadable "**Basic Database**" by the function "**Export as Basic DB**" in the menu of the starting page (**Fig. 9**). This saves a copy of the prepared empty database as **tourcount0_dl_Tourname.db** within the app’s files directory "**Documents/TourCount**".

The app data directory “**Documents/TourCount”** is created during the first app call. When uninstalling TourCount you will not loose your data as this directory remains untouched.

The basic database does not contain any count data, comments, or inspection-specific metadata. It just stores the structure of the set up DB and serves as a template for further tours recordings in the region. It should be renamed suitably, e.g. tourcount0_dl_region xyz.db.
Note: The file name of a **Basic DB** must always start with the string “**tourcount0**“ ("**tourcount**" with a trailing "**0**").

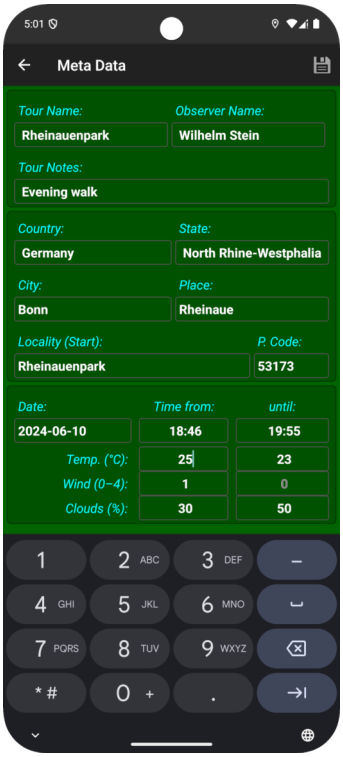


Fig. 5: Edit Metadata page

After importing a Basic DB, all that remains to prepare for a field survey is to enter the survey-specific metadata (date, start time, temperature, wind, and cloud cover).

The naming convention for filenames of the exported TourCount files:
- Basic DB: **tourcount0_dl_Tourname.db**
- Current DB: **tourcount_dl_StartDate_Tourname.db**
- Result: **Tour_dl_StartDate_Tourname_osc.xlsx** or **.csv**
- Species list: **species_dl_Tourname_CurrentDate.csv**

With
- **blue**: fixed string parts,
- **dl**: data language of local species names ("**de**", "**en**", "**es**", "**fr**", "**it**"),
- **Tourname**: tourname (filtered for filename conformity),
- **StartDate**: date of the tour start (**YYYYMMDD-hhmm**), or if not specified, the current storage date,
- **osc**: output sorting code ("**n**" = species names, "**c**" = species codes) and
- **CurrentDate**: current storage date (**YYYYMMDD-hhmm**).

3. Usage

After the app launches and your location is detected (after about 10 sec.), a message about the first locality found is displayed and, if applicable, with a message indicating that the detected locality is outside the European fauna area (as a reminder that only European species are supported).

Start with "**Edit Metadata**". Enter the specific metadata of the tour (**Fig. 5**). Date and time can be entered by tapping the relevant field. If a different date or time is to be entered, these fields can be pressed longer for an input dialog.
To suppress unreal values for temperature, wind or clouds, plausibility checks are done when saving and refuse saving until suitable values are entered. The percentage value entered for cloud cover is rounded to the nearest ten when saved. Store the data and finish the function with the
💾 **button**.

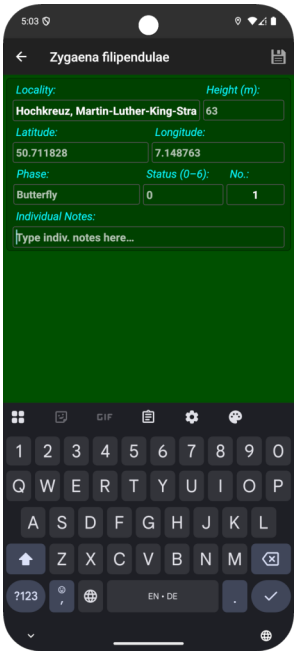


Fig. 6: Edit Individual Data page

Continue with "**Counting**" (**Fig. 2**). Select the species by clicking the species line to scroll down the species list. To count just click the (**+**)-**button** of the corresponding category (♂|♀, ♂, ♀, pupa, larva, egg). The counter will increment and the **Individuals page** (**Fig. 6**) opens.
Location info, latitude, longitude, height as well as date and time will be added automatically. The location may be edited, condition rating state (0 - 6 with 0 = undefined, 1 = very good, 6 = very bad) and multiple counts may be applied here. The **"Individual Notes"** field lets you select a common individual note from a pop-up list and/or edit it.

With the 💾 **button** accept the entered individual data and go back to the counting page. To go back without saving the individual data, use the ⬅-Button. This also resets the count to its previous value.

The (**-**)-**buttons** on the Counting page allow for corrections. Mind that the (**-**)-button reduces the individuals list in a last-in-first-out-mode for the respective category.

The **"Species Notes"** field on the counting page lets you enter notes or select a common species note from a pop-up list with a single click.
If you enter a "**0**" in this field, the species will also be listed in the results later, despite a count of 0.

To move back one page you generally can use the ⬅-button. To make sure to save changed content you should go back by clicking the respective ⊕, 🗑 or 💾 **button**.

Before finishing TourCount after a tour you should complete the metadata and save the current reckonings as an MS Excel file by "**Export Results (xlsx)**" from the menu of the starting page to **Tour_dl_StartDate_Tourname_osc.xlsx** in the directory **Documents/TourCount**. The filename contains info about the data language, the tour name and the start date with time. If no start date is specified in the metadata, the current date is included.

The Counting Results page (**Fig. 7 and 8**) can be opened on the starting page with "**Show results**" or the 👁 **icon**. It shows all the registered data orderly arranged. Beneath the metadata it shows the totals followed by a list of all counted species with their individual records sorted by the recorded time.

When calling up pages with large scroll lists, the response of the app can be somewhat delayed, as extensive calculations are running in the background. If the message "**TourCount not responding**" appears, please reply with **"Wait"**.

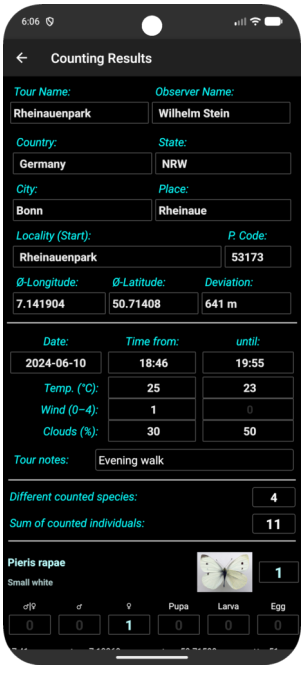


Fig. 7: Results page (head)

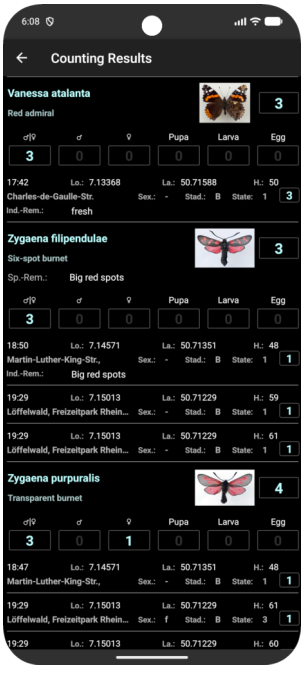


Fig. 8: Results page (continued)

4. Further Functions

The main menu on the starting page (**Fig. 9**) has Settings, Reset, Import, Export, Help and Info functions.

In "**Settings**" (**Fig. 10**) you may adapt the look an feel in some aspects to your wishes, e.g. sorting order of lists and data export, left-/right-hand counting page, sounds and haptic button feedback as well as GPS-based location assistance

Here are a few explanations regarding the latter:
Once GPS synchronization is complete and the initial address query to the "**Nominatim Reverse Geocoding**"[1] service is successful, the initial location details of the starting point (**country, state, ZIP code, city, town, locality**) are obtained and then automatically incorporated into the metadata when editing them. For counted individuals, the current locality name besides coordinates and height is automatically added to the individual's data record.

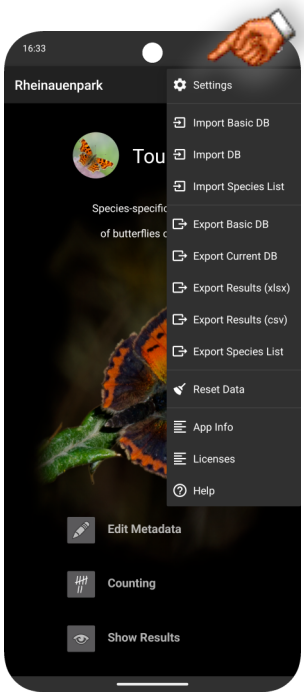


Fig. 9: Main Menu

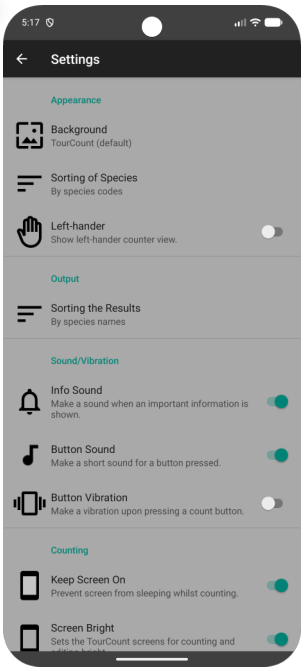


Fig. 10: Settings (partly)

The "**Address request interval**" can be set between 10 and 30 seconds (default: 20 seconds). The free Nominatim service should be used only as needed locally and as sparingly as possible.
To strike a balance between performance and power consumption, the "**Min. Distance for GPS Update**" can be set between 2 and 50 m (default: 10 m) and the "**GPS Polling Interval**" between 2 and 20 seconds (default: 5 seconds).

For preparing a new tour you may use "**Reset Data**" to reset the tour-specific metadata and count data. Alternatively you may import a stored Basic DB, another TourCount-DB or a species list.

However, after that, the current locality is not recognized again until about 10 seconds later, at which point it is set as the starting locality. Only after the corresponding pop-up message appears can the locality's metadata be automatically inserted on the metadata page. If the metadata page is accessed too quickly, the current location data will be missing. This data would only be added when the metadata page is accessed at a later time, but then probably with a wrong locality.

Android-specifically, TourCount stores the data in the same, equally named SQLite DB file in the the app's own internal storage area. As this file cannot be read or changed directly by the user, importing or exporting the data from/to files in a user accessible storage area is necessary.

Starting with app version 3.7.4, DB files are exported with a field for identifying the data language for local species names (German, English, French, Italian, or Spanish). For imported DB files created prior to this update, the correct data language should be selected via "Settings" to ensure proper processing.

With "**Import Basic DB**" you can select and import an existing Basic DB file that was exported as a template for a regional tour and begins with "**tourcount0**". However, after the import, it takes about 10 seconds for the current locality to be recognized again.
Note: The file name must always begin with the exact character string "**tourcount0**" (e.g., "tourcount0_en_KottenForest.db"), otherwise the file cannot be imported as a Basic DB.

When **exporting data**, the tour name (if it exists in the metadata) will be inserted into the name of the exported file. Here, any characters that violate the filename conventions are removed as necessary.

[1] For Reverse Geocoding (to produce address info from GPS coordinates) the service of Nominatim from OpenStreetMap is used. A valid own email address is necessary for durably reliable queries of address data and to exclude abuse. The mail address will be treated confidentially and will only be used to contact you in case of service problems. For more info see https://wiki.openstreetmap.org/wiki/Nominatim.

With “**Export Basic DB**”, the internal DB (without walk-specific data but with general metadata and the current species list) can be exported as the Basic-DB file “**tourcount0_dl_Tourname.db**” into **Documents/TourCount**. This is reasonable if permanent changes have been made to the count list, such as adding new or removing species (see “**2. Setting up**”).

With “**Import DB**,” an exported or appropriately named TourCount DB containing count data can be selected and loaded into the app. However, after the import, it takes about 10 seconds for the current locality to be recognized again.

Note: The file name must always begin with the exact character string “**tourcount_**” (e.g., “tourcount_en_KottenForest-20250815.db”), otherwise the file cannot be imported as a normal DB.

“**Export Current DB**” writes a copy of the entire DB with all currently contained data as “**tourcount_dl_StartDate_Tourname.db**” into **Documents/TourCount**. If exists, the start date and time of the tour get inserted into the filename. Otherwise, the current date gets inserted. If a tour name exists, it is inserted into the filename but eventually gets cleaned up to remove characters that violate file naming conventions.

The function “**Import Species List**” lets you select and import an available species list. The metadata is retained during import, but all counting data is deleted. If the TransektCount app is installed, the exported species lists can be imported mutually.
Starting with app version 3.7.2, the species list has been updated to include the data language. The available data languages are German, English, French, Italian, and Spanish. When local species names were available, they have been included for the species listed in the app.
However, after the import, it takes about 10 seconds for the current locality to be recognized again.

Note: The file name of a species list must always begin with “**species_**”, otherwise the file cannot be imported.

With **"Export Species List"** the current species list without any other data can be exported as a CSV file (CSV = Comma Separated Values text file) **"species_dl_Tourname_CurrentDate.csv**”.

If you imported a species list that was exported from an app version earlier than 3.7.2, you should set its data language in Settings and then re-export the list. After that, the data language will be automatically detected and set in the future.

The function “**Export Results (xlsx)**” writes the counting results with the tour metadata into the formatted Excel spreadsheet file “**Tour_dl_StartingDate_Tourname_osc.xlsx**” to “**Documents/TourCount**”. If exists, the start date and time of the tour get inserted into the filename. Otherwise, the current date gets inserted.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	**Counting List: Rheinauenpark**				**Observer: Wilhelm Stein**						Species sorted by species names			
2											Individuals sorted by registered time			
3	**Country**	**State**		**P. Code**	**City**		**Place**		**Locality (Start)**					
4	Germany	NRW		53173	Bonn		Rheinaue		Rheinauenpark					
5														
6	**Date**		**Time**	**Temp. (°C)**	**Wind (0–4)**	**Clouds (%)**					**Tour Notes**			
7	2024-06-10	from:	18:46	25	1	30					Evening walk			
8		until:	19:55	23	0	50								
9														
10	**Species Name**	**English name**		**Spec. Code**	**Counts ♂|♀**	**Counts ♂**	**Counts ♀**	**Counts Pu.**	**Counts Ca.**	**Counts Eg.**	**Spec. Notes**			
11	Macroglossum stellatarum	Hummingbird hawk-moth		06843							0			
12	Pieris rapae	Small white		06998			1							
13	Vanessa atalanta	Red admiral		07243	3									
14	Zygaena filipendulae	Six-spot burnet		03998	3	1					Big red spots			
15	Zygaena purpuralis	Transparent burnet		03974	3	1	1							
16	**Diff. Species:**	**4**		**Totals:**	**9**	**2**	**2**				**Sum Total:**	**13**		
17														
18	**Individuals**	**Locality**		**Counts**	**Longitude**	**Latitude**	**Uncertainty (m)**	**Height (m)**	**Date**	**Time**	**Sexus**	**Phase**	**Status (0–6)**	**Indiv. Notes**
19	Pieris rapae	Charles-de-Gaulle-Str.		1	7.13368	50.71588	47	51	2024-06-10	17:41	f	Butterfly	5	old
20	Vanessa atalanta	Charles-de-Gaulle-Str.		3	7.13368	50.71588	56	50	2024-06-10	17:42	-	Butterfly	1	fresh
21	Zygaena purpuralis	Martin-Luther-King-Str.		1	7.14571	50.71351	55	48	2024-06-10	18:47	-	Butterfly	1	
22	Zygaena purpuralis	Martin-Luther-King-Str.		1	7.14571	50.71351	55	48	2024-06-10	18:47	m	Butterfly	-	
23	Zygaena filipendulae	Martin-Luther-King-Str.		1	7.14571	50.71351	45	48	2024-06-10	18:50	-	Butterfly	1	photo
24	Zygaena purpuralis	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	50	61	2024-06-10	19:29	f	Butterfly	3	
25	Zygaena purpuralis	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	46	60	2024-06-10	19:29	-	Butterfly	1	
26	Zygaena filipendulae	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	47	59	2024-06-10	19:29	-	Butterfly	1	
27	Zygaena purpuralis	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	46	61	2024-06-10	19:29	-	Butterfly	1	
28	Zygaena filipendulae	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	45	61	2024-06-10	19:29	-	Butterfly	1	
29	Zygaena filipendulae	Löffelwald, Freizeitpark Rheinaue		1	7.15013	50.71229	45	61	2024-06-10	19:29	m	Butterfly	-	
30														
31					**Longitude**	**Latitude**	**Uncertainty (m)**							
32			**Average Coordinates:**		7.141904	50.71408	641							

Fig. 11: Exported XLSX table opened in LibreOffice Calc

Fig. 11 shows an Excel Results file in the LibreOffice Calc application.

The function “**Export Results (csv)**” writes the counting results with the tour metadata into the preformatted, comma separated .csv textfile “**Tour_dl_StartingDate_Tourname_osc.csv**” to “**Documents/TourCount**”. If exists, the start date and time of the tour get inserted into the filename. Otherwise, the current date gets inserted.

The exported table can be sorted in advance (by species or codes) using the result sorting option in the TourCount settings, depending on your preference for easier data entry into a central monitoring web page like **https://www.falterfunde.de/platform/s4y/falterfunde/index.do**.

The exported result files could be viewed and, to a limited extent, edited in a spreadsheet app such as **Collabora** (open source, free, based on LibreOffice). It is available in the Collabora Office Store (an additional package source in the F-Droid app store), in the Play Store, or directly at **https://www.collaboraoffice.com/downloads/**).

The exported .xlsx or .csv result files can also be transferred to a PC for further processing (see **6.1 Tips**) and loaded or imported into a spreadsheet program (tested with MS Excel, LibreOffice Calc, and Softmaker PlanMaker).

IT-affine users may also transfer the exported database (.db) files to a PC and examine or edit them there with a free tool like "**DB Browser for SQLite**" (**https://sqlitebrowser.org**). It lets you edit them manually or per SQL script. Some useful example SQL scripts are published in the docs directory of the GitHub TourCount project site **https://github.com/wistein/TourCount/tree/master/docs**.

Under "**App Info**" you find the email address of the author and the history of the app development.

Under **"Licenses"** you find the license notes for code and figures.

The option menu on the counter page has a **photo function** to start the camera as quickly as possible without leaving TourCount.

A TourCount-specific message can also be sent there by **share function** via SMS or e-mail.

Optionally, the counter view is temporarily turned off by means of the **proximity sensor** when the phone is pocketed or closely held to the body. This saves energy, prevents accidental entries and brings the app back immediately when it is used normally again.

The app shows several **short pop-up messages** in 4 categories:

- **Error** (e.g. with a database).
- **Warning** (e.g. in the event of an invalid entry).
- **Note** (e.g. if an action might last longer).
- **Info** (e.g. after a successful process).

5. Installation Hints

5.1 Install from F-Droid store (released version, recommended)

Get the TourCount app without docs or species list examples but with updates from the F-Droid app store:

App: **https://f-droid.org/en/packages/com.wmstein.tourcount**

Note on F-Droid:
Getting apps from F-Droid is at least as secure as getting them from the Google Play Store.
In contrast to the Play Store, all apps are also checked for privacy protection and compiled by F-Droid itself.
If an app does not meet all of F-Droid's requirements with regard to undesirable features, it is noted.

The source codes of the F-Droid apps are published and licensed with an open source license.

5.2 Get the app from the author's GitHub project pages (current beta status)

App: **https://github.com/wistein/TourCount/tree/master/apk**

Download the **TourCount_xxx.apk** file from the GitHub page into the Downloads directory.
On the smartphone activate the Android option "Unknown source" in "Settings -> Security" before Installation. Then use the filemanager to install the app by clicking on the .apk file in the download folder.

Important: Finally deactivate the option "Unknown source"!

5.3 Basic setting up the app

After installation start TourCount once to create the necessary data directory **Documents/TourCount**.
Then you may get the documentation and relevant species lists from the author's GitHub pages.

Docs: **https://github.com/wistein/TourCount/tree/master/docs**

Copy these from the download folder into the new directory Documents/TourCount.
Now you can restart the app and finish setting it up.

In addition, the GitHub documentation directory mentioned above contains some sample SQL scripts for editing .db files (SQLite database files) for Insiders.

5.4 Notes on updates

Unfortunately, the two installation methods mentioned above are not mutually compatible for updates, since although the .apk files are generated from identical source code, they are signed with different keys.

In the case of major version jumps with functional additions, structural changes may have been made in the internal database of the app. After such a change, the database version is incremented. This is recognized by the app and the currently used DB is automatically upgraded internally.

All sample databases are included in the current version.
However, a DB file in its current version may not be backward compatible with previous app versions.

6 Annex

6.1 Tips

Transferring data between smartphone and PC

Connect the smartphone to the PC using a USB cable. In the smartphone notifications find „**Charging this device via USB**“, tab for more options and select „**Use USB for file transfer**“.

In Windows Explorer or under Linux in Nautilus, Nemo or similar file manager the smartphone is now shown with its technical ID. In the "**Internal shared memory**" area, the "**Documents/TourCount**" directory can be read and written to for data exchange.

6.2 Glossary

CSV file:
Comma-separated values file. Text-based file format for exchanging data in tabular form (e.g. for importing TourCount result data into spreadsheet programs).

File directories of TourCount:
The public app-specific files directory (for DB files and exported CSV files) is:

"Documents/TourCount"

Data stored here can also be read by other apps. The data is not automatically deleted when the app is uninstalled.

The previous app-specific directory for the DB files:

"Android/data/com.wmstein.tourcount/files"

is no longer used from TourCount version 3.4.5, as data stored here cannot be read by other apps in newer Android versions and was deleted when the app got uninstalled.

GitHub:
Is a file hosting site for software development projects including version control. It is free of charge for Open Source Projects. Eponymous was the version control and source code management system Git. Run by GitHub, Inc. from San Francisco, USA. Since December 2018 the company belongs to Microsoft. According to Microsoft GitHub will remain an independent platform.

Numbering scheme according to Karsholt/Razowski:
The entomologists O. Karsholt and J. Razowski developed a numbering scheme for all European butterfly species, that among others is used by the German Lepiforum. According to that numbering scheme TourCount uses Codes for the identification of species. However, this limits the utilization of TourCount to European faunal areas, as there is no adequate global scheme.

Open Source:
Source code of software, which can be edited and used publicly. Open source software can mostly be used free of charge and does not contain propriately licensed or closed source elements.

6.3 References

TourCount

TourCount project:

The repository of the TourCount project is situated on **https://github.com/wistein/TourCount**.
It contains all published files with source code, configuration of the Android Studio Development platform, documentation and installable APK files.

TourCount Documents:

You find documents, example species lists, SQL scripts for manipulating TourCount DBs and other information under **https://github.com/wistein/TourCount/tree/master/docs**.

TourCount-App:

The released version can be installed from
https://f-droid.org/en/packages/com.wmstein.tourcount/.

TransektCount

TransektCount is the complementary Android app to support transect counters in nature preserving projects according to the Butterfly Monitoring Scheme methodology in Europe. It allows a species-specific recording of individuals in all stages per transect section.

TransektCount project:

The repository of the TransektCount project is situated on **https://github.com/wistein/TransektCount**.
It contains all published files with source code, configuration of the Android Studio Development platform, documentation and installable APK files.

TransektCount documents:

You find documents, example species lists, SQL scripts for manipulating TransektCount DBs and other information under **https://github.com/wistein/TransektCount/tree/master/docs**.

TransektCount app:

The released version can be installed from
https://f-droid.org/en/packages/com.wmstein.transektcount/.